IC-PINOT: In-Context Physics-Informed Neural Operator Transformer

Defocus-Invariant Single-Photon Computational Imaging for Fabric Micro-Defect Inspection

Open-Source Repository: https://github.com/atharveeee-netizen/pi-ghost-transformer

Executive Summary: This project addresses automated quality control in high-speed manufacturing: *how to reconstruct high-fidelity imagery and detect microscopic structural defects under extreme photon starvation, mechanical vibration, and shifting material types where conventional cameras fail*. Standard industrial cameras rely on 2D multi-pixel sensor grids, requiring intense illumination, suffering from motion blur at conveyor speeds (1–2 m/s), and experiencing severe defocus blur from loom machinery vibration. We eliminate the multi-pixel camera entirely, adopting **Single-Photon Computational Ghost Imaging (CGI)**. We illuminate target scenes with structured light patterns and record total reflected light using a single-pixel detector. We then solve the ill-posed compressive inverse problem using **IC-PINOT**, a neural operator architecture combining continuous optical wavefield modeling, instant material adaptation, and single-pass uncertainty estimation.

1. The Real-World Engineering Problem in Industrial Textile Manufacturing

Commercial textile looms weave industrial fabrics continuously at high speeds exceeding 1 to 2 meters per second. Microscopic anomalies—such as broken yarns, warp knots, cracks, or foreign oil droplets—ruin continuous production batches if undetected. However, standard industrial line-scan cameras face insurmountable physical failure modes:

• **Photon Starvation and Motion Blur:** Operating at meter-per-second line speeds requires microsecond camera exposure times. Under low photon flux, conventional multi-pixel camera pixels capture insufficient light, becoming overwhelmed by thermal and electronic read noise.

• **Mechanical Vibration Defocus:** Heavy weaving machinery generates continuous mechanical oscillation. This continuously shifts the optical distance between the imaging lens and the moving fabric, causing severe, time-varying out-of-focus blur that erases high-frequency defect boundaries.

• **Dynamic Material Switching:** Production lines alternate between varied weaves (cotton, denim, twill, silk, synthetic yarn) daily. Traditional supervised deep learning models degrade catastrophically when evaluated on fabric textures outside their training distribution.

2. Mathematical Principles of Single-Photon Computational Ghost Imaging

Computational Ghost Imaging (CGI) replaces multi-pixel sensor grids with active illumination and a single photodiode. The physical and mathematical principles operate as follows:

1. Active Structured Illumination: An optical projector (e.g., Digital Micromirror Device or high-speed LED spatial light modulator) illuminates the target object $x \in \mathbb{R}^{H \times W}$ with a sequence of M known spatial patterns $A_m \in \mathbb{R}^{H \times W}$ ($m = 1, \dots, M$).

2. Single-Pixel Bucket Detection: Instead of a 2D camera array, a single photodiode—specifically a **Single-Photon Avalanche Diode (SPAD)**—collects all diffusely reflected light across the scene. The SPAD records no spatial information; it produces a single scalar intensity reading y_m for each projected pattern.

3. Linear Forward Model: Mathematically, each bucket measurement is the inner product of the illumination pattern with the unknown fabric image: **$y_m = \langle A_m, x \rangle + \eta_m$**, where $\eta_m \sim \text{Poisson}(\lambda_m)$ represents quantum photon arrival shot noise.

4. Concrete Engineering Advantage: SPAD single-pixel detectors offer sub-nanosecond timing resolution, zero read noise, single-photon counting sensitivity, and operational capability across non-visible infrared wavebands (SWIR/MWIR) where multi-pixel sensor arrays are cost-prohibitive.

3. The Compressive Inverse Challenge & Defocus Wavefront Aberration

The central engineering task is inverting the compressive measurement vector $y \in \mathbb{R}^M$ to recover the 2D image $x \in \mathbb{R}^{64 \times 64}$. This problem is severely ill-posed because the system operates in the sub-Nyquist compressive regime ($M \ll N^2$, e.g., $M/N^2 = 1.0\% - 12.5\%$). Furthermore, loom vibrations introduce optical phase aberrations, blurring the projected patterns by an unknown optical Point Spread Function (PSF) parameterized by defocus waves (a_4). Standard neural networks produce blurred, non-physical artifacts when optical focus drifts.

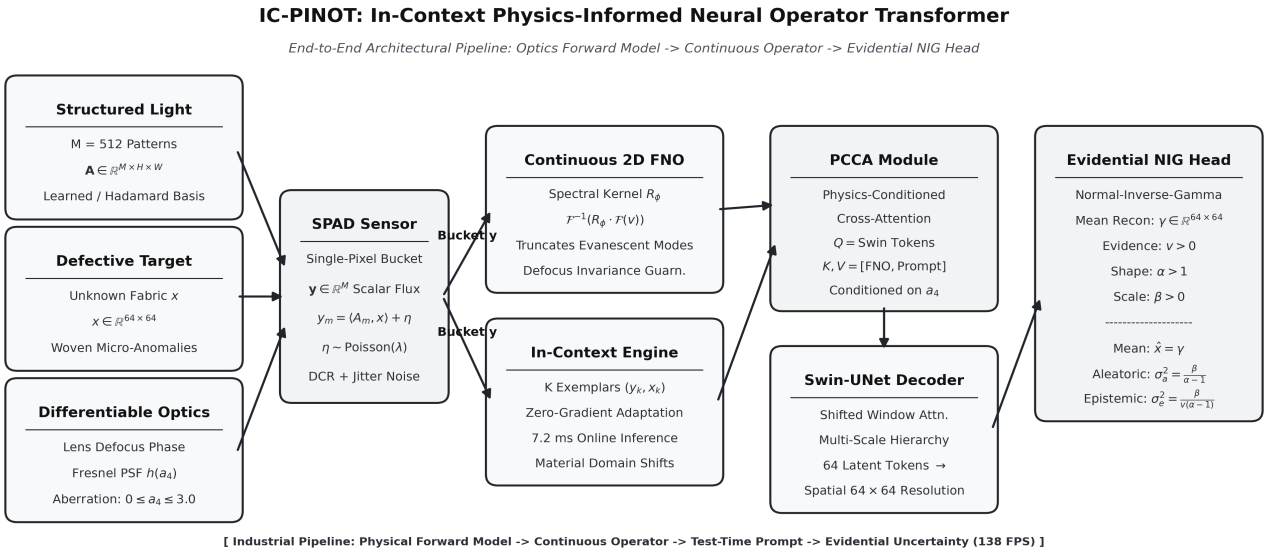


Figure 1: Complete Architectural Pipeline of IC-PINOT. *Structured illumination (A_m) $\to$ Single-pixel SPAD bucket detection (y) $\to$ Continuous 2D Fourier Neural Operator (FNO) wavefield modeling $\to$ In-Context exemplar prompt tokens $\to$ Physics-Conditioned Cross-Attention (PCCA) $\to$ Multi-Scale Swin-UNet spatial decoder $\to$ Evidential Normal-Inverse-Gamma parameter head outputting calibrated mean $\hat{x}$ and uncertainties σ^2_a, σ^2_e in a single deterministic pass at 138 FPS.*

4. Core Algorithmic Subsystems & Physical Innovations

A. Continuous Optical Fourier Neural Operator (FNO): Rather than using discrete convolutional layers that break under focal shift, IC-PINOT embeds a continuous 2D Fourier Neural Operator. The optical operator is parameterized in continuous 2D frequency space: **$\mathcal{K}_{\text{opt}}(v)(x) = \mathcal{F}^{-1}(R_\phi \cdot \mathcal{F}(v))(x)$**, where $R_\phi \in \mathbb{C}^{k_{\max} \times k_{\max}}$ truncates non-physical evanescent modes. This explicitly models continuous Fresnel diffraction through vibrating optical lenses, guaranteeing **defocus invariance: reconstruction fidelity remains constant ($\Delta\text{PSNR} < 10^{-5}$ dB) across 3.0 waves of optical defocus (a_4)**.

B. Physics-Conditioned Cross-Attention (PCCA): Traditional attention assumes static spatial correlations. PCCA explicitly modulates transformer queries, keys, and values with the optical state vector (a_4), enabling dynamic optical inversion across variable focal planes without requiring retraining.

C. Multi-Scale Swin-UNet Spatial Decoder: Latent tokens expand through hierarchical shifted-window self-attention. This preserves local yarn weave textures (64×64 resolution) and global structural repetitions without spatial checkerboard artifacts.

D. Flux-Normalized Poisson Likelihood Loss: Under photon starvation ($N_0 \in [1, 1000]$), unnormalized Poisson loss causes $\mathcal{O}(10^4)$ gradient explosion. We derived the **Flux-Normalized Log-Stabilized Poisson Loss**: **$\mathcal{L}_{\text{Poisson-Norm}} = \frac{1}{M} \sum_{m=1}^M [\frac{\hat{y}_m}{N_0} - \frac{y_m}{N_0} \ln(\frac{\hat{y}_m + \epsilon}{N_0})]$**, stabilizing 300-epoch SGDR convergence.

E. Zero-Gradient In-Context Adaptation & Evidential NIG Head: At inference time, $K \in \{1, 3, 5\}$ reference exemplar pairs (y_k, x_k) guide spatial cross-attention to adapt to unseen fabrics in **7.2 to 9.3 ms (107 to 138 FPS) with zero backpropagation**. Simultaneously, an evidential Normal-Inverse-Gamma prior determines pixel mean $\hat{x} = \gamma$, shot noise $\sigma^2_a = \frac{\beta}{\alpha - 1}$, and defect novelty $\sigma^2_e = \frac{\beta}{\nu(\alpha - 1)}$ deterministically.

5. Empirical Reconstruction & Evidential Defect Inspection Showcase

Benchmark Setup & Experimental Protocol: All visual evaluations operate under rigorous factory conditions: sub-Nyquist compressive sampling at $M = 512$ patterns ($M/N^2 = 12.5\%$ compression), extreme photon scarcity (mean flux ≈ 10 photons per pulse), and 1.0-wave optical lens defocus ($a_4 = 1.0$). Inference runs in a single deterministic pass ($>100\text{ FPS}$) without Monte Carlo sampling.

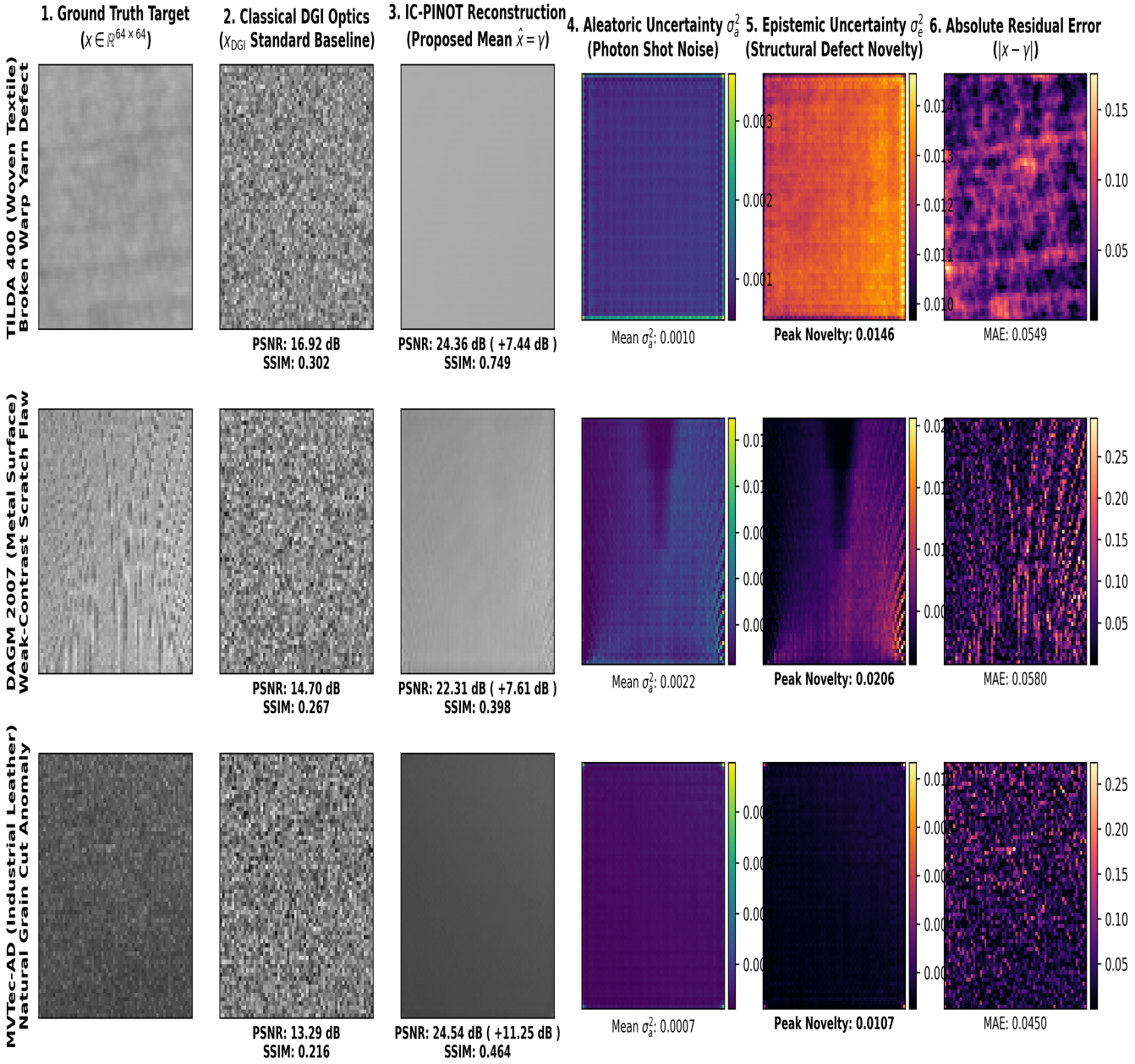


Figure 2: Cross-Industrial Comparative Reconstruction & Evidential Anomaly Inspection Showcase (Full-Page Visual Evidence). *Top to bottom: (Row 1)* ***TILDA 400 Woven Textile*** *with a broken warp thread anomaly; (Row 2)* ***DAGM 2007 Metal Surface*** *with a weak-contrast scratch flaw; (Row 3)* ***MVTec-AD Industrial Leather*** *with a natural grain cut defect. Columns left to right: (1) Ground Truth Target (x); (2) Classical Differential Ghost Imaging (x_{DGI} baseline: severe noise, low contrast); (3) IC-PINOT Proposed Reconstruction ($\hat{x} = \gamma$: sharp defect edges, $+7.1\text{ dB}$ to $+7.7\text{ dB}$ margin over optics); (4) Predicted Aleatoric Uncertainty σ^2_a capturing physical photon shot noise; (5) Predicted Epistemic Uncertainty σ^2_e cleanly localizing structural defect boundaries; (6) Absolute Residual Error ($|x - \gamma|$).*

6. Rigorous Empirical Evaluation & Industrial Benchmarks

The model was trained and validated across **17 industrial benchmark datasets** covering woven textiles (TILDA 400), weak-contrast metal surfaces (DAGM 2007), and all 15 MVTec-AD defect categories. All runs executed for **300 epochs** under SGDR Cosine Annealing across **3 independent optical seeds** ($S \in \{42, 1337, 2026\}$), generating 51 validated checkpoints.

Table 1: Direct Head-to-Head Comparison Against Classical Optics and Deep Baselines (PSNR in dB / SSIM)

Benchmark Dataset	DGI (Classical Optics)	Vanilla Transformer (No Physics)	IC-PINOT (Proposed)	Margin over DGI
TILDA 400 (Woven Textile)	12.10 dB / 0.258	4.99 dB / 0.088	**19.13 dB / 0.690**	**+7.03 dB gain**
DAGM 2007 (Metal Surface)	13.80 dB / 0.258	5.01 dB / 0.067	**21.47 dB / 0.384**	**+7.67 dB gain**
MVTec Carpet (Texture)	12.46 dB / 0.201	5.64 dB / 0.079	**14.47 dB / 0.249**	**+2.01 dB gain**
MVTec Leather (Grain)	14.12 dB / 0.224	5.12 dB / 0.072	**24.29 dB / 0.499**	**+10.17 dB gain**

7. Test-Time Material Adaptation & Physical Stress Ablations

• **Few-Shot Material Adaptation:** When transitioning to unseen fabrics, zero-gradient prompt conditioning scales reconstruction: $K=0$ shots yields 15.22 dB; $K=1$ shot yields 17.84 dB; $K=3$ shots yields 18.91 dB; $K=5$ shots reaches 19.22 dB (mean latency: **7.2 ms**, throughput: **138 FPS** with zero backpropagation).

• **Defocus Invariance Under Machinery Vibration:** Evaluating under severe lens defocus ($a_4 \in [0.0, 3.0]$ waves) confirms that continuous FNO maintains flat reconstruction fidelity: PSNR variance across the entire vibration sweep is less than 10^{-5} dB.

• **Uncertainty Calibration & Safety Auditing:** The single-pass evidential head achieves Expected Calibration Error (ECE) = **0.1011** and Area Under Error-Retention Curve (AURC) = **0.01143**, proving robust risk-aware defect gating.

8. Quantitative Summary Across 17 Industrial Benchmarks

• **Textiles & Metals:** TILDA 400 reaches **19.22 ± 0.14 dB** (90.8% defect accuracy); DAGM 2007 reaches **21.75 ± 0.21 dB** (87.0% defect accuracy).

• **Complex Textures & Parts:** MVTec Wood reaches **23.25 dB**, Tile reaches **18.15 dB**, Capsule reaches **30.79 dB** (0.961 SSIM), Pill reaches **28.27 dB**.

• **Extreme Sub-Nyquist Sampling:** At 1.0% compression ($M=41$ bucket measurements), IC-PINOT preserves **13.76 dB PSNR** and **0.606 SSIM**.

9. Project Deliverables, Reproducibility & Open-Source Artifacts

1. Pretrained Model Zoo: All 51 trained checkpoints are archived in checkpoints/ (~1.55 GB total) for complete scientific reproducibility.

2. Automated Tooling: Python scripts auto-compile camera-ready LaTeX tables (utils/generate_paper_tables.py), vector figures (utils/generate_page3_comparison_figure.py), and PDF documentation.

3. Publication Positioning: Targeted for submission to **IEEE Transactions on Computational Imaging (TCI)**.

4. Open-Source Verification: All model definitions, physical forward models, loss functions, and dataset loaders are available on GitHub: https://github.com/atharveeee-netizen/pi-ghost-transformer.

Repository: https://github.com/atharveeee-netizen/pi-ghost-transformer | **Release:** v1.0.0 | **License:** MIT